

Joint peak shaving and frequency regulation strategy for energy storage stations cluster considering battery dispatchable capacity

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ABSTRACT

This paper proposes a joint response strategy for peak shaving (PS) and frequency regulation (FR) in energy storage (ES) stations cluster to address uneven response capacity distribution, significant unit status variations, and insufficient sustained operation capacity in regional power grids. The strategy first predicts the state of health (SOH) of energy storage units, followed by a two-stage coordinated optimization process. In the first stage, a station-stage correction coefficient for dispatchable capacity is introduced, considering residual joint response capacity. A power allocation model is subsequently developed to minimize operational costs while responding to joint PS and FR commands, ensuring cost-effective coordination among stations. In the second stage, unit-stage optimization adjusts available capacities based on SOH predictions, introduces state of charge (SOC) weight coefficients and interval-based recovery targets, and constructs a power redistribution model to minimize SOC recovery deviation, ensuring SOC recovery and energy balancing. Simulation results validate that the proposed strategy enhances the response accuracy of multiple ES stations to joint service commands and improves operational continuity and economic performance.

1. Introduction

The rapid integration of renewable energy sources, such as wind and photovoltaic power, presents significant challenges to the safe and stable operation of power grids. The volatility and intermittency of wind and solar energy can cause sharp frequency fluctuations, threatening grid stability. Additionally, the load peak-to-valley difference continues to widen due to variations in wind speed and solar irradiance [1]. Energy storage (ES) systems, characterized by their rapid response and high-precision bidirectional regulation capabilities, are increasingly recognized as essential flexible resources for supporting grid ancillary services. Energy storage systems can rapidly address frequency deviations for frequency regulation (FR) and temporally shift energy to facilitate peak shaving (PS) and valley filling, thereby enhancing overall system stability. The need to optimize ES valuation while maintaining robust PS and FR performance has become a central concern in both theoretical studies and practical implementations [2].

A review of existing theoretical research reveals various mature strategies for ES participation in single auxiliary services, such as PS or FR. A constraint model for ES output under FR commands has been

developed in reference [3] within a scheduling framework, which enhances tracking performance by specifying constraints on FR responses. Further studies have proposed a two-layer control structure to enhance ES response accuracy and frequency stability, as shown in [4]. Reference [5] has also presented a model for assessing the state of charge (SOC) safety of ES in response to FR, which supports stability during frequency fluctuations. Meanwhile, performance-based cost analyses of FR have been examined, and a comparison of FR costs based on service quality and tracking accuracy requirements has been proposed [6]. The comparison results indicate that FR provides greater benefits for battery operation. Furthermore, the responsiveness to FR in multi-energy storage systems has been investigated in [7]. The aforementioned studies have demonstrated promising results in FR, response accuracy, and system stability. In the field of PS research, a system-level ES response model using a multi-scenario net load has been constructed, providing a theoretical basis for ES utilization in PS under large-scale renewable energy integration [8]. It has been argued that deep PS is more suitable for the new type of power system, based on the theoretical basis for PS [9]. A multi-resource response PS linear planning model has been further developed to effectively smooth the load, reducing the peak-to-valley difference and improving grid operation efficiency [10].

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Nomenclature

Indices and Set

t	Index of time slots
T	Set of time slots
j	Index of energy storage (ES) units
J	The number of ES units within an ES station
i	Index of ES stations
I	The number of ES stations in the region

Variable

$P_{i,t}^{\text{bat}}, P_{i,j,t}^{\text{bat}}$	The dispatchable capacity of the ES station i and its internal unit j
$P_{i,t}^{\text{cha}}, P_{i,t}^{\text{dis}}, P_{i,j,t}^{\text{cha}}, P_{i,j,t}^{\text{dis}}$	The dispatchable charging/discharging capacities of the ES station i and its internal unit j
$P_{i,t}^{\text{PS}}, P_{i,t}^{\text{FR}}$	The power of the ES station i in response to peak shaving (PS)/frequency regulation (FR) commands
$P_{i,t}^{\text{PS,cha}}, P_{i,t}^{\text{FR,dis}}$	The charging/discharging power of the ES station i in response to PS commands
$P_{i,t}^{\text{FR,cha}}, P_{i,t}^{\text{FR,dis}}$	The charging/discharging power of the ES station i in response to FR commands
$E_i^{\text{bat}}, E_{i,j}^{\text{bat}}$	The current capacity of the ES station i and its internal unit j
$\Delta S_{i,j,t}$	The state of charge (SOC) recovery deviation of ES unit
$S_{i,j,t}, S_{i,j,t-1}$	The actual SOC values of the ES unit at the current and previous moment
$\alpha_{i,t}^{\text{con,cha}}, \alpha_{i,t}^{\text{con,dis}}$	The dispatchable capacity charging/discharging correction coefficients of the ES station i
$D_{i,t}^{\text{con,cha}}, D_{i,t}^{\text{con,dis}}$	The normalized dispatchable capacity charging/discharging correction coefficients
$D_{i,t}^{\text{con}}$	The dispatchable capacity correction coefficients of the ES station i
$P_{i,t}^{\text{rem,cha}}, P_{i,t}^{\text{rem,dis}}$	The residual charging/discharging response capacities of the ES station i
$P_{i,t}^{\text{rem}}$	The residual response capacity of the ES station i
$\rho_{i,j,t}^{\text{cha}}, \rho_{i,j,t}^{\text{dis}}$	The charging/discharging SOC weight coefficients of the

$\omega_{i,j,t}^{\text{cha}}, \omega_{i,j,t}^{\text{dis}}$	unit j in the ES station i
$\omega_{i,j,t}$	The normalized charging/discharging SOC weight coefficients
$C_{i,t}^{\text{inv}}, C_{i,t}^{\text{loss}}, C_{i,t}^{\text{life}}, C_{i,t}^{\text{om}}, C_{i,t}^{\text{cost}}$	The initial construction cost, energy loss cost, life-cycle depreciation cost, maintenance cost and total cost of ES station i

Constant

Δt	The time slot
$\varepsilon_t^{\text{PS}}, \varepsilon_t^{\text{FR}}$	The PS command and FR command
c_i^{cap}	The unit capacity cost of the ES station i
T_i^{float}	The float life of the ES station i
r	The discount rate
s	The maximum dispatchable quantity within a day
c_i^{bat}	The unit power cost of the ES station i
c_i^{ep}	The electricity selling price of ES station i
c_i^{om}	The maintenance cost factor of ES station i
N_i	The number of charge-discharge cycles of ES station i
λ	The dispatchable capacity margin for a joint response
$\eta_i^{\text{cha}}, \eta_i^{\text{dis}}$	The charging/discharging efficiency of the ES station i
$E_i^{\text{rated}}, P_i^{\text{rated}}$	The rated capacity and rated power of the ES station i
$E_{i,j}^{\text{rated}}, P_{i,j}^{\text{rated}}$	The rated capacity and rated power of the unit j in ES station i
$\underline{S}, \bar{S}$	The minimum and maximum values of the target SOC recovery range
$S_{i,j}^{\text{min}}, S_{i,j}^{\text{max}}$	The minimum and maximum SOC of the unit j in ES station i
$u_t^{\text{cha}}, u_t^{\text{dis}}$	The charging/discharging signals in response to joint commands
$u_t^{\text{PS,cha}}, u_t^{\text{PS,dis}}$	The charging/discharging signals in response to the PS command
$u_t^{\text{FR,cha}}, u_t^{\text{FR,dis}}$	The charging/discharging signals in response to the FR command

Although the aforementioned work has effectively enhanced PS response and adapted to load fluctuations, its focus on single service scenarios (PS or FR) has limited its applicability to the multi-task flexible service potential of ES systems under resource cross-use across different regulation tasks in actual operation. Moreover, it has not addressed the relationship between the time scales of PS and FR. Consequently, scholars have begun exploring joint optimization strategies for PS and FR to address these limitations. Reference [11] has proposed a joint control strategy for PS and FR, leveraging ES's fast-response capabilities to achieve superlinear gains and simplifying operations via real-time threshold control algorithms. Building on this foundation, Reference [12] has explored a techno-economic evaluation model across multiple time scales by integrating long-term PS and short-term FR, providing a comprehensive framework for balancing economic and technical objectives in multi-time scale operations. Furthermore, Reference [13] has developed a two-layer optimization model for PS and FR, focusing on minimizing cost loss within system safety constraints. Similarly, Reference [14] has extended this approach to multiple ES systems, developing a more advanced two-layer optimization model to optimize cost-safety synergies and enhance the multi-energy storage response. These studies have achieved dual assurance of revenue and reliability by establishing economic indicators, incorporating degradation and unit output costs, and aligning with the health state of ES. These findings

have highlighted that the multi-task synergistic ability of multi-energy storage systems in complex environments significantly outperforms single technologies, offering a new perspective on multi-objective optimization [15]. Though these studies have partly addressed the limitations of single auxiliary services and shown some economic benefits of synergy, most still focus on individual ES systems, lacking a framework to coordinate multiple ES systems in regional grids. This gap stems from the complex interplay between modeling methods and time scale influences, impeding a unified approach for multi-energy storage coordination.

Current research in power system scheduling, especially regarding the integration of multiple ES stations, has primarily focused on ES aggregation and task decomposition. Although these studies have been innovative in task hierarchy division and control architecture design, they have lacked a cooperative and optimized scheduling mechanism for power system scheduling. This is mainly because their objectives have been predominantly commercial operation or single-point control. Consequently, the scheduling process has typically treated the ES station as a single entity for power response, ignoring state factors such as SOC differences and capacity degradation among storage units within the ES station. However, the responsiveness of ES systems also depends on the state of health (SOH) and SOC of individual battery units. Since SOH cannot be measured directly, accurate prediction is crucial for reliable

system dispatch. As highlighted in [16], future ES reliability assessments must incorporate SOH prediction. To support optimal scheduling and account for unit-level differences, this study adjusts the available capacity of each unit based on predicted SOH. Reference [17] has summarized the commonly used data-driven methods for predicting the SOH of Li-ion batteries, highlighting the technological advantages of LSTM and BiLSTM. Although some studies have recognized the importance of SOH estimation, they have often used empirical thresholds or mean value approximations, which fail to reflect state differences between ES units. To enhance model viability and precision, a CNN-BiLSTM-Attention method has been employed to construct an SOH prediction model for each ES unit [18]. This approach provides detailed health state estimation through multi-dimensional feature extraction and time series degradation modeling, and adjusts the available ES capacity accordingly, supporting subsequent optimal scheduling.

In summary, significant progress has been made in optimizing ES systems for PS and FR. A solid foundation has been established in multi-service coupling, scheduling optimization, and economic efficiency. However, existing research lacks a systematic framework for co-dispatching multiple ES stations, modeling unit state differences, and incorporating health state awareness in regional grids. This paper focuses on the optimal scheduling of multiple electrochemical ES stations for joint PS and FR in regional grids. A strategy for joint PS and FR response of ES stations cluster, considering battery dispatchable capacity, is proposed to address coordination and optimization challenges. The strategy employs a CNN-BiLSTM-Attention model to predict the SOH of ES units, thereby adjusting their available capacities accordingly. Coordinated optimization is then performed from the station to the unit stage. This optimization aims to enhance the stability and efficiency of the system's operation. The main contributions of this paper are summarized as follows:

- 1) A two-stage scheduling framework is proposed to incorporate SOH predictions and coordinate PS and FR demands, achieving global optimal dispatch from the station level to the unit level.
- 2) At the station level, the strategy integrates residual response capacity and a dispatchable capacity correction coefficient to minimize the operational cost and allocate joint regulation tasks efficiently.
- 3) At the unit level, output optimization uses SOH-based capacity correction along with SOC interval constraints and weight coefficients to maintain energy balance and support state recovery.

The rest of the paper is organized as follows. Section 2 proposed the SOH prediction method. Section 3 proposes a two-stage scheduling framework and optimization models. Section 4 describes the solution process and evaluation metrics. Section 5 presents case studies and validates the strategy. Section 6 concludes with key contributions and future work.

2. SOH prediction method for ES units with dispatchable capacity considerations

This study focuses on the joint response of multiple ES stations in a regional power grid to PS and FR demands. Each ES station consists of multiple units whose performance degrades over time due to cycling and aging. Consequently, their dispatchable capacity may no longer reflect actual operational limits, impacting scheduling reliability. To address this, assessing the current SOH of each unit is essential for accurate dispatchable capacity quantification. Accordingly, this study presents an SOH prediction model based on the CNN-BiLSTM-Attention method for internal storage units before introducing the proposed two-stage optimization strategy for coordinated joint response [18]. This model underpins subsequent capacity correction and optimal scheduling. The proposed model integrates a CNN, BiLSTM network, and attention mechanism to accurately evaluate ES unit SOH. The model architecture includes five layers: input, CNN, BiLSTM, Attention, and output. By

combining CNN feature extraction, BiLSTM temporal modeling, and attention-based weighting, the model significantly enhances SOH prediction accuracy. The SOH prediction process includes three stages: data preprocessing stage, model training stage, and model testing stage, as shown in Fig. 1.

3. A two-stage response strategy for ES stations cluster under PS and FR demands

3.1. The framework for a two-stage response strategy

Building on the SOH prediction results presented in Section 2, this study develops a two-stage response strategy for coordinating multiple ES stations within a regional power grid with integrated renewable energy sources. As shown in Fig. 2, the regional control center utilizes updated dispatchable capacities to prioritize renewable energy consumption and adjust ES outputs in response to joint PS and FR commands. In the first stage, the optimization considers the response cost, dispatchable capacity correction, and residual capacity of ES stations during joint operation, with the objective of minimizing the overall joint response cost. The outcome is the dispatch output assigned to each ES station for the joint command, which serves as the input for the second optimization stage. This stage incorporates SOC weight coefficients for each unit to minimize SOC recovery deviations.

3.2. Station-stage strategy

The objective of the station stage is to minimize the operational cost of ES stations in response to joint commands. This cost considers both the dispatchable capacity correction coefficient and the residual response capacity during joint operation. Energy storage stations with lower joint response costs and higher residual capacities are prioritized for fulfilling the joint command. Conversely, ES stations with higher joint response costs and lower residual capacities are assigned minimal response responsibilities. The results serve as inputs for the SOC optimization of ES units in the subsequent phase. Firstly, the stage introduces two concepts: $P_{ij,t}^{\text{bat}}$ and $P_{i,t}^{\text{bat}}$. The relationship equation between them is presented below.

$$\begin{cases} P_{ij,t}^{\text{bat}} = P_{ij,t}^{\text{dis}} - P_{ij,t}^{\text{cha}} \\ P_{i,t}^{\text{bat}} = P_{i,t}^{\text{dis}} - P_{i,t}^{\text{cha}} \end{cases} \quad (1)$$

The regional ES station responds to dispatch commands from the power scheduling center, including PS command ξ_t^{PS} and total automatic generation control (AGC) FR command ξ_t^{FR} , by generating four control signals: $u_t^{\text{PS,cha}}$, $u_t^{\text{PS,dis}}$, $u_t^{\text{FR,cha}}$, and $u_t^{\text{FR,dis}}$. Specifically, $u_t^{\text{PS,dis}} = 1$ when $\xi_t^{\text{PS}} > 0$, and $u_t^{\text{PS,cha}} = 1$ when $\xi_t^{\text{PS}} < 0$. Otherwise, both are set to 0. Similarly, $u_t^{\text{FR,dis}} = 1$ when $\xi_t^{\text{FR}} > 0$, and $u_t^{\text{FR,cha}} = 1$ when $\xi_t^{\text{FR}} < 0$.

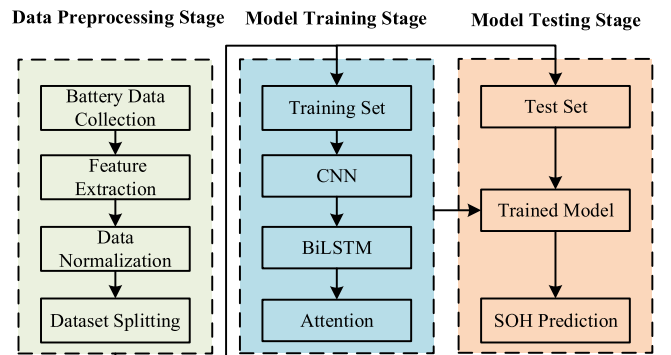


Fig. 1. The prediction process of SOH for ES units based on CNN-BiLSTM-Attention.

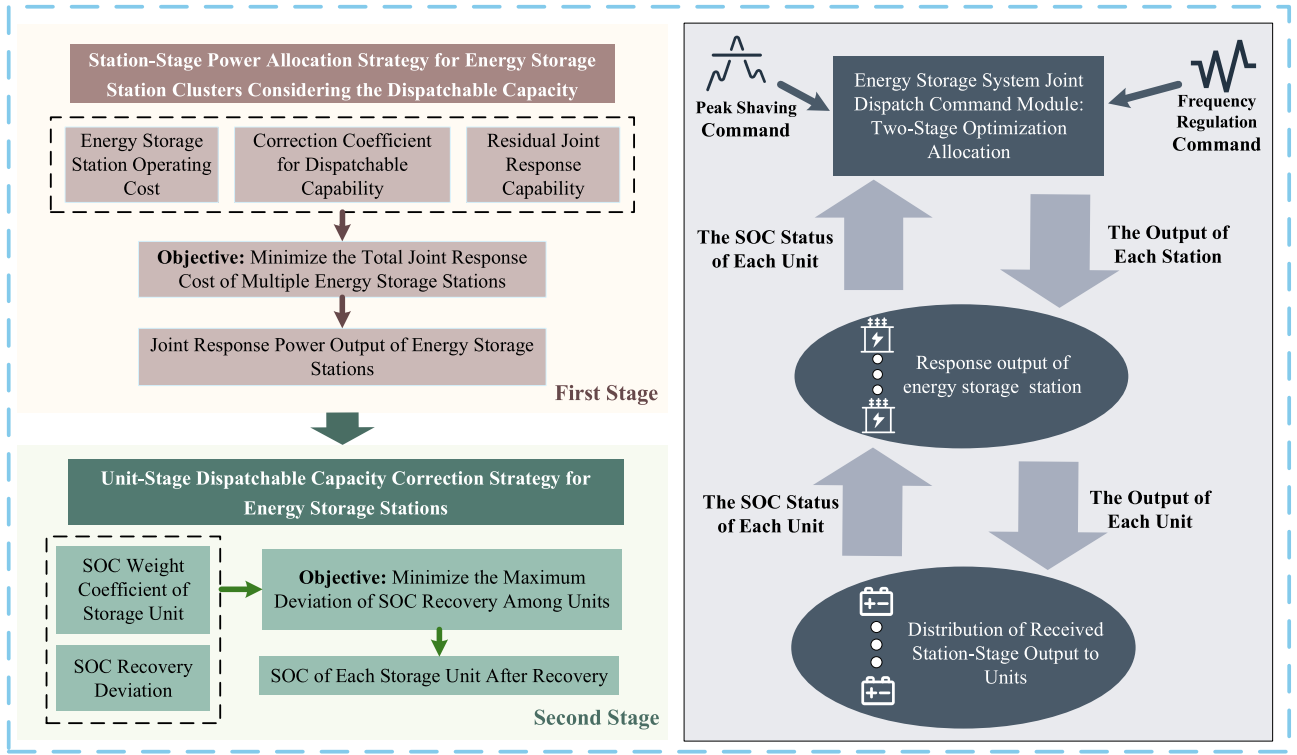


Fig. 2. The framework for a two-stage response strategy for ES Stations cluster.

After defining the joint response capacity components and corresponding signals for each dispatch command, the joint response cost model for each ES station is simplified. The predicted SOH is used to estimate available capacity via the capacity method, and this is included in the cost calculation. The total cost mainly includes initial construction, energy loss, and life-cycle depreciation, and further accounts for maintenance costs and battery replacement expenses [19]. Given the relatively short operating horizon, battery replacement expenses are subsumed under maintenance costs, and the overall cost structure is formulated in Eqs. (2)–(6).

$$E_{ij}^{\text{bat}} = \text{SOH} \cdot E_{ij}^{\text{rated}} \quad (2)$$

$$C_{i,t}^{\text{inv}} = \frac{c_i^{\text{cap}} \sum_{j=1}^J E_{ij}^{\text{bat}}}{365s} \frac{r(1+r)^{T_{i,t}^{\text{float}}}}{(1+r)^{T_{i,t}^{\text{float}}} - 1} \quad (3)$$

$$C_{i,t}^{\text{loss}} = c_i^{\text{ep}} \left[P_{i,t}^{\text{dis}} (1/\eta_i^{\text{dis}} - 1) + P_{i,t}^{\text{cha}} (1 - \eta_i^{\text{cha}}) \right] \Delta t \quad (4)$$

$$C_{i,t}^{\text{life}} = \frac{c_i^{\text{bat}} \left(P_{i,t}^{\text{cha}} \eta_i^{\text{cha}} + P_{i,t}^{\text{dis}} / \eta_i^{\text{dis}} \right) P_i^{\text{rated}} \Delta t}{2 \sum_{j=1}^J E_{ij}^{\text{bat}} N_i} \quad (5)$$

$$C_{i,t}^{\text{om}} = c_i^{\text{om}} C_{i,t}^{\text{inv}} \quad (6)$$

The total cost $C_{i,t}^{\text{cost}}$ of the joint operation is expressed in Eq. (7).

$$C_{i,t}^{\text{cost}} = C_{i,t}^{\text{inv}} + C_{i,t}^{\text{loss}} + C_{i,t}^{\text{life}} + C_{i,t}^{\text{om}} \quad (7)$$

This study extends that framework by defining a dispatchable capacity correction coefficient and a residual response margin to represent coordinated operation. The correction coefficient is inversely related to residual capacity: lower residual discharge or charge capacity results in a higher correction coefficient for discharging or charging, respectively. In joint operations, when an ES station has a high cost but sufficient

residual capacity, the discharging coefficient is minimized to maintain output flexibility. Conversely, when residual capacity is constrained and substantial charging margin remains, the charging coefficient is reduced to enhance charging efficiency. These principles form the basis for the model in Eq. (8).

$$\begin{cases} \alpha_{i,t}^{\text{con,cha}} = \sum_{j=1}^J E_{ij}^{\text{bat}} (S_{ij,t-1} - S_{ij}^{\text{min}}) / (S_{ij}^{\text{max}} - S_{ij}^{\text{min}}) \\ \alpha_{i,t}^{\text{con,dis}} = \sum_{j=1}^J E_{ij}^{\text{bat}} (S_{ij}^{\text{max}} - S_{ij,t-1}) / (S_{ij}^{\text{max}} - S_{ij}^{\text{min}}) \end{cases} \quad (8)$$

The coefficients are then normalized, as defined in Eq. (9).

$$\begin{cases} D_{i,t}^{\text{con,cha}} = \alpha_{i,t}^{\text{con,cha}} / \sum_{i=1}^I \alpha_{i,t}^{\text{con,cha}} \\ D_{i,t}^{\text{con,dis}} = \alpha_{i,t}^{\text{con,dis}} / \sum_{i=1}^I \alpha_{i,t}^{\text{con,dis}} \end{cases} \quad (9)$$

This paper defines joint charging and discharging command signals, where the u_t^{cha} (charging) and u_t^{dis} (discharging) commands are taken as the maximum values of the corresponding PS and FR commands. To ensure mutually exclusive operation, a constraint $u_t^{\text{cha}} + u_t^{\text{dis}} \leq 1$ is imposed. The joint signals are then used to construct the dispatchable capacity correction coefficient $D_{i,t}^{\text{con}}$, as shown in Eq. (10).

$$D_{i,t}^{\text{con}} = u_t^{\text{cha}} D_{i,t}^{\text{con,cha}} + u_t^{\text{dis}} D_{i,t}^{\text{con,dis}} \quad (10)$$

The station stage achieves optimal dispatchable capacity allocation by accounting for residual response capacity and minimizing joint response cost. The objective function integrates the correction coefficient and cost terms, as defined in Eq. (11).

$$F_1 = \min \sum_{i=1}^I D_{i,t}^{\text{con}} C_{i,t}^{\text{cost}} \quad (11)$$

Following the construction of the objective function, the

dispatchable capacity for joint response is constrained by the residual response capacity $P_{i,t}^{\text{rem}}$ of the ES station i . This residual capacity is determined by the current SOC of ES units in the station i and the current capacity. Control is achieved through the joint charging and discharging command signals, as expressed in Eqs. (12) and (13).

$$P_{i,t}^{\text{rem}} = u_t^{\text{cha}} P_{i,t}^{\text{rem,cha}} + u_t^{\text{dis}} P_{i,t}^{\text{rem,dis}} \quad (12)$$

$$\begin{cases} P_{i,t}^{\text{rem,cha}} = \sum_{j=1}^J \min \left[P_{ij}^{\text{rated}}, E_{ij}^{\text{bat}} (S_{ij,t-1}^{\text{max}} - S_{ij,t-1}) / (\eta_i^{\text{cha}} \Delta t) \right] \\ P_{i,t}^{\text{rem,dis}} = \sum_{j=1}^J \min \left[P_{ij}^{\text{rated}}, E_{ij}^{\text{bat}} (S_{ij,t-1} - S_{ij,t-1}^{\text{min}}) \eta_i^{\text{dis}} / \Delta t \right] \end{cases} \quad (13)$$

To accurately describe the dispatchable charging and discharging capacity of the ES station i in response to joint commands, the total output is further decomposed into two components: one corresponding to the PS command and the other corresponding to the FR command. On this basis, to better reflect the coordination and allocation behavior of ES stations under dual-command response, the relationship between PS and FR power allocation is further formulated in Eq. (14).

$$P_{i,t}^{\text{x}} = P_{i,t}^{\text{x,dis}} - P_{i,t}^{\text{x,cha}}, \text{ x} \in \{\text{PS}, \text{FR}\} \quad (14)$$

After determining the residual response capacity of the ES stations participating in joint operations and sorting their power in response to PS and FR commands, this study examines the relationship between the total residual response capacity of the regional ES stations cluster and the PS and FR commands when formulating power output constraints. The study subsequently classifies the constraints on dispatchable capacity optimization for ES stations responding to joint commands into three categories, defined as follows:

1) When the condition $\lambda \sum_{i=1}^I P_{i,t}^{\text{rem}} \geq |\xi_t^{\text{PS}}| + |\xi_t^{\text{FR}}|$ holds, it implies that the total residual joint response capacity of the regional ES stations cluster is sufficient to meet the current joint response command while maintaining a specified dispatchable capacity margin. Additionally, the charging and discharging power constraints of ES, as described in previous work [20]. The corresponding constraint is formulated as follows:

$$\begin{cases} \sum_{i=1}^I P_{i,t}^{\text{x,cha}} = -u_t^{\text{x,cha}} \xi_t^{\text{x}} \\ \sum_{i=1}^I P_{i,t}^{\text{x,dis}} = u_t^{\text{x,dis}} \xi_t^{\text{x}} \end{cases}, \text{ x} \in \{\text{PS}, \text{FR}\} \quad (15)$$

2) When the condition $|\xi_t^{\text{FR}}| \leq \lambda \sum_{i=1}^I P_{i,t}^{\text{rem}} < |\xi_t^{\text{PS}}| + |\xi_t^{\text{FR}}|$ holds, it implies that the total residual coordinated response capacity is insufficient to meet the joint response command but adequate for the individual FR command. The corresponding constraint is formulated as follows:

$$\begin{cases} \sum_{i=1}^I P_{i,t}^{\text{FR,dis}} = u_t^{\text{FR,dis}} \xi_t^{\text{FR}} \\ \sum_{i=1}^I P_{i,t}^{\text{FR,cha}} = -u_t^{\text{FR,cha}} \xi_t^{\text{FR}} \end{cases} \quad (16)$$

$$\begin{cases} \sum_{i=1}^I P_{i,t}^{\text{PS,cha}} = u_t^{\text{PS,cha}} \left(\lambda \sum_{i=1}^I P_{i,t}^{\text{rem}} + \xi_t^{\text{FR}} \right) \\ \sum_{i=1}^I P_{i,t}^{\text{PS,dis}} = u_t^{\text{PS,dis}} \left(\lambda \sum_{i=1}^I P_{i,t}^{\text{rem}} - \xi_t^{\text{FR}} \right) \end{cases} \quad (17)$$

3) To address situations where the total residual coordinated response capacity may be insufficient to meet the joint response command, each ES station responds according to its available capacity. If the FR command still cannot be fully satisfied, that is $\lambda \sum_{i=1}^I P_{i,t}^{\text{rem}} < |\xi_t^{\text{FR}}|$, the system prioritizes fulfilling the FR command to the greatest extent possible. The corresponding constraint is formulated as follows:

$$\begin{cases} P_{i,t}^{\text{FR,cha}} = \lambda u_t^{\text{FR,cha}} P_{i,t}^{\text{rem,cha}} \\ P_{i,t}^{\text{FR,dis}} = \lambda u_t^{\text{FR,dis}} P_{i,t}^{\text{rem,dis}} \end{cases} \quad (18)$$

3.3. Unit-stage strategy

This study addresses the SOC optimization stage by allocating the charging and discharging response power of each ES unit within the station, based on the assigned joint response capacity $P_{i,t}^{\text{bat}}$ and the SOC status of the internal units. This allocation is designed to improve the stability and flexibility of coordinated response operations within the ES stations cluster. In contrast to the approach in [21], which focuses on a single SOC recovery value, this study introduces a novel interval-based method. The control objective is relaxed from point-based recovery to maintaining the SOC within a designated interval $[\underline{S}, \bar{S}]$, formulated as an out-of-interval deviation. Correction is applied only when the SOC exceeds the interval bounds, thereby enhancing the robustness of the SOC optimization process for ES systems. The recovery deviation is expressed in Eq. (19).

$$\Delta S_{ij,t} = \max(\underline{S} - S_{ij,t}, 0) + \max(S_{ij,t} - \bar{S}, 0) \quad (19)$$

The unit stage incorporates SOC weight coefficients to allocate the joint response capability among ES units within each ES station based on their individual SOC values. This allocation prioritizes charging for units with lower SOC and discharging for units with higher SOC, thereby balancing the SOC values of ES units. The specific formulation is expressed in Eq. (20).

$$\begin{cases} \beta_{ij,t}^{\text{cha}} = E_{ij}^{\text{bat}} (S_{ij,t-1}^{\text{max}} - S_{ij,t-1}) / (S_{ij,t}^{\text{max}} - S_{ij,t}^{\text{min}}) \\ \beta_{ij,t}^{\text{dis}} = E_{ij}^{\text{bat}} (S_{ij,t-1} - S_{ij,t-1}^{\text{min}}) / (S_{ij,t}^{\text{max}} - S_{ij,t}^{\text{min}}) \end{cases} \quad (20)$$

To prevent the objective function from being excessively large, the coefficients are normalized into $\omega_{ij,t}^{\text{cha}}$ and $\omega_{ij,t}^{\text{dis}}$ using a method similar to that in Eq. (9). Additionally, the SOC weight coefficients are governed by the joint charging and discharging signals, with their expressions defined as follows:

$$\omega_{ij,t} = u_t^{\text{cha}} \omega_{ij,t}^{\text{cha}} + u_t^{\text{dis}} \omega_{ij,t}^{\text{dis}} \quad (21)$$

The unit stage minimizes SOC deviations beyond specified limits, guiding ES units to operate within the target range, as shown in Eq. (22).

$$F_2 = \min \sum_{i=1}^I \sum_{j=1}^J \omega_{ij,t} \Delta S_{ij,t} \quad (22)$$

The SOC optimization of ES units is subject to constraints, as defined in previous work [17], with the actual capacity of the ES units corrected using the SOH incorporated within these constraints.

4. Solution method and effect evaluation

4.1. Solution method

To illustrate the execution process of the proposed optimization strategy, the corresponding flowchart is shown in Fig. 3. The process begins with SOH prediction of ES units, followed by calibration of their available capacity. The procedure then proceeds to a two-stage optimization framework. In the first stage, the dispatchable capacity correction coefficient is introduced, and power is allocated to joint PS and FR commands at the station level based on operating cost and residual response capacity. The second stage focuses on ES units within each station, using SOC weight coefficients and SOC recovery optimization to enable intra-station power redistribution and state recovery.

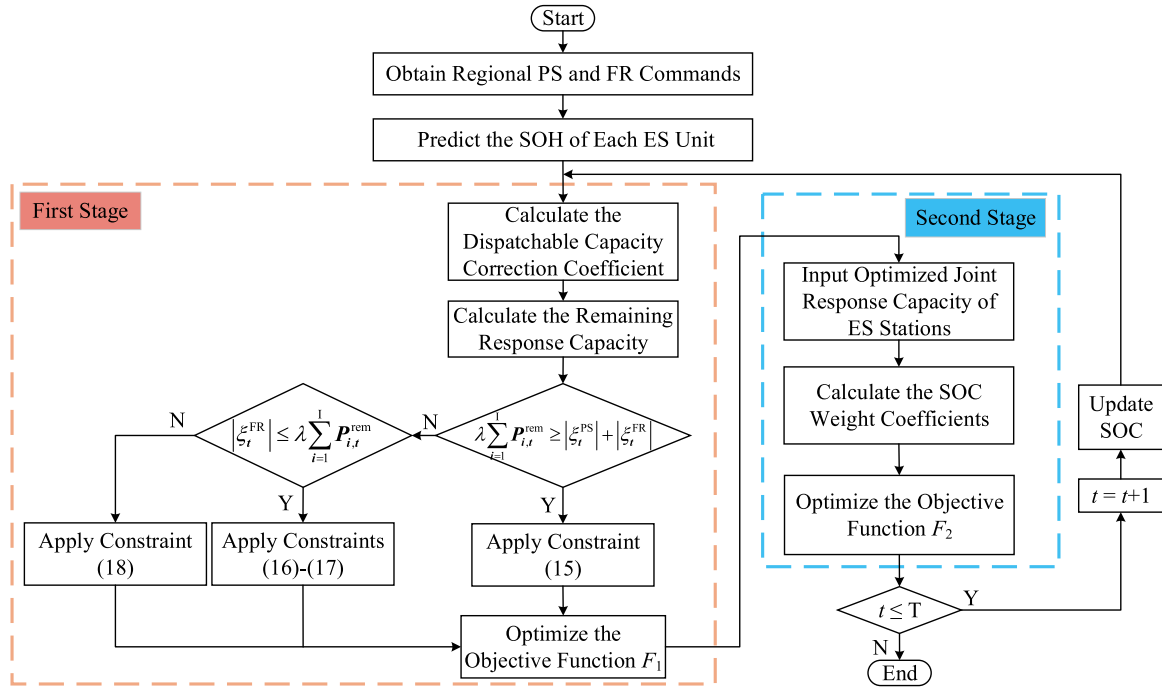


Fig. 3. Flowchart of two-stage response strategy for the ES stations cluster considering battery dispatchable capacity.

4.2. Evaluation indicators for strategy effectiveness

Two evaluation indicators are proposed to assess the effectiveness of the two-stage optimization strategy. These indicators are detailed below [22]:

- 1) The deviation indicator for the ES response to PS and FR outputs

$$\varepsilon_{\text{index}}^{\text{x, RMSE}} = \sqrt{\frac{1}{T} \sum_{t=1}^T \left(\sum_{i=1}^I P_{i,t}^x - \varepsilon_t^x \right)^2}, x \in \{\text{PS, FR}\} \quad (23)$$

- 2) The deviation indicator for SOC recovery of each ES unit

$$\varepsilon_{\text{index},i}^{\text{SOC}} = \sqrt{\frac{1}{TJ} \sum_{t=1}^T \sum_{j=1}^J \left[\max \left(0, \underline{S} - S_{ij,t}, S_{ij,t} - \bar{S} \right) \right]^2} \quad (24)$$

5. Case studies

5.1. Simulation setup

To evaluate the technical and economic performance of the proposed strategy, simulation analyses are conducted using actual PS and FR command data from a regional power grid. The response intervals are set to 5 min for PS commands and 1 min for FR commands, with a joint response dispatchable capacity margin of 0.95. The feed-in tariff for ES stations in the target region is 643.77 CNY/MWh. The case studies involve three electrochemical ES stations, each with a maintenance cost factor of 5 % and a discount rate of 8 %, and comprising six ES units. Detailed configuration parameters for each ES station are provided in Table 1, and initial SOC values for ES units are listed in Table 2. Due to differences in the actual cycle life of individual ES units, their SOH values are predicted using the CNN-BiLSTM-Attention method introduced in Section 2. Given the varying cycle life across all ES units, the predicted SOH values for all units are summarized in Table 3 to provide a clear overview of the results.

5.2. Results and analysis

The response of the regional ES stations cluster to the proposed joint

Table 1

The detailed configuration parameters of each ES station.

Parameter Name	ES Station 1	ES Station 2	ES Station 3
Charging/Discharging Efficiency	0.85	0.95	0.9
Rated Power/MW	3 × 6	3 × 6	3 × 6
Rated Capacity/MWh	3 × 6	3 × 6	3 × 6
Unit Power Cost (CNY/kW)	2500	1500	1500
Unit Capacity Cost (CNY/(kWh))	5000	3000	900
Float Life (Years)	12	10	10
Cycle Life (Cycles)	3000	2000	2500
Maximum SOC	0.9	0.9	0.9
Minimum SOC	0.1	0.1	0.1

Table 2

The initial SOC values of ES units in each ES station.

ES Station	1	2	3	4	5	6
1	0.70	0.80	0.75	0.80	0.85	0.70
2	0.20	0.15	0.25	0.15	0.20	0.30
3	0.50	0.55	0.60	0.55	0.50	0.60

Table 3

The predicted SOH values of ES units in each ES station.

ES Station	1	2	3	4	5	6
1	0.91	0.93	0.90	0.92	0.91	0.94
2	0.90	0.91	0.92	0.90	0.91	0.92
3	0.93	0.93	0.91	0.92	0.95	0.92

command strategy is shown in Fig. 4. The strategy enhances coordinated response to PS and FR commands, enabling more effective power allocation, especially under high demand. During the third joint response period (91–96 min), the FR command is effectively tracked while a partial response to the PS command is maintained. In the fourth period (131–136 min), sustained discharging reduces the residual capacity, leading to a temporary FR tracking deviation between 132 and 134 min. At 135 min, response to the FR command resumes, alongside partial PS

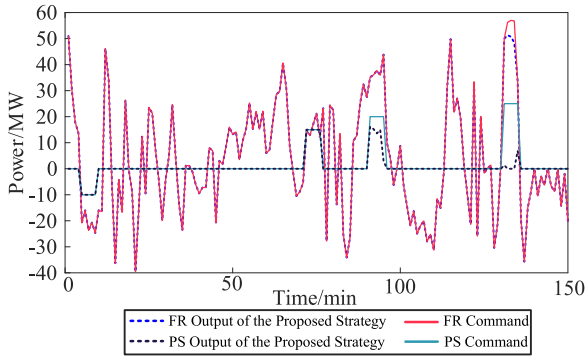


Fig. 4. Joint command response of ES stations cluster under the proposed strategy.

response, indicating the recovery in joint capability. Overall, the strategy improves recovery performance, bidirectional regulation, and dynamic scheduling flexibility, demonstrating clear advantages in coordinated dispatch.

5.3. Discussion

5.3.1. Comparisons of different strategies

To verify the effectiveness of the proposed two-stage optimization strategy, four comparative strategies are designed:

Strategy A: Adopts the mean allocation strategy directly proposed in [23], serving as a benchmark for comparison.

Strategy B: Minimizes the total joint response cost without the dispatchable capacity correction coefficient and second-stage optimization.

Strategy C: Applies only the first-stage optimization.

Strategy D: Implements the full two-stage optimization proposed in this study.

As shown in Fig. 5, the response performance is presented from 80 min onward, since all strategies satisfy the joint command before this time. However, during the third joint response period, Strategy A exhibits FR shortages at 93 and 95 min and fails to provide any PS response. Strategies B and C also lack sufficient energy for effective response. Strategy B shows continuous FR shortages from 91 to 96 min

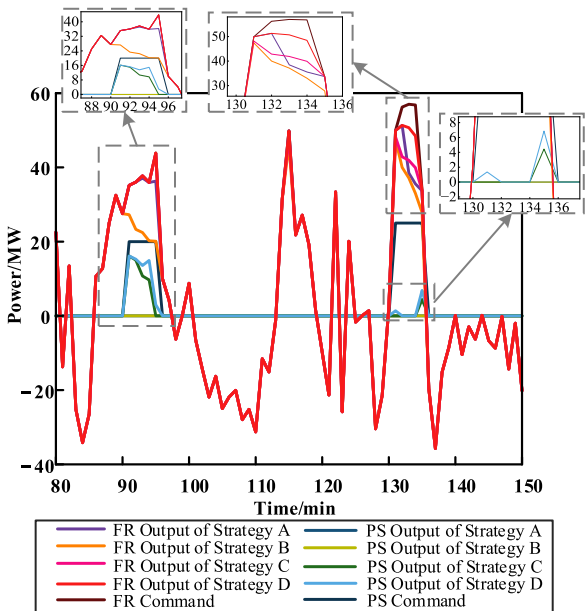


Fig. 5. Joint command response of ES stations cluster under different strategies.

and also struggles to meet the PS command. Strategy C maintains adequate FR response from 91 to 94 min, enabling partial PS response after meeting FR demand, but exhibits FR shortages from 95 min onward. In the fourth period, Strategy A delivers slightly higher FR output than Strategies B and C but still provides no PS response. Strategy B fails to meet both commands at 131 min due to depleted energy and prioritizes FR over PS. Strategy C also falls short on FR but delivers higher FR output than Strategy B. At 135 min, both Strategy C and Strategy D satisfy the FR command and respond to the PS command, with Strategy D providing the highest PS response. Overall, Strategy D outperforms the others in terms of both FR and PS response capabilities.

The performance of the four strategies is analyzed under several metrics. Various ES parameters indicate that ES station 1 has the highest unit joint response cost, while ES station 3 has the lowest. As shown in Table 4, Strategy D achieves a cumulative total FR output of 2668.384 MW and a total PS output of 196.118 MW, both higher than those of Strategies A, B, and C. Moreover, Strategy D achieves the lowest PS output deviation, measured at 4.587. Furthermore, it records the smallest FR output deviation, reaching 0.945, thereby outperforming the other three strategies.

Strategy A, which adopts dynamic proportional power allocation, performs slightly better than Strategy B. However, its operating cost is higher, and its response effectiveness and economic efficiency are inferior to those of Strategies C and D. Strategies C and D demonstrate better coordination and efficiency compared to Strategy B, mainly due to the inclusion of dispatchable capacity correction coefficients in their optimization models. This design allows ES stations with higher initial discharge capacities, such as ES station 1, to take on more response tasks in the early stage. As shown in Fig. 6(c) and Fig. 6(d), ES station 2 responds more effectively to charging instructions during the first 50 min under Strategies C and D. In the third response phase, this results in improved performance for Strategies C and D, while Strategy B shows the weakest response.

1) Impact of Dispatchable Capacity Correction Coefficients on Strategy Performance

The correction coefficients help coordinate between unit cost and available capacity. During discharge instructions, all stations respond. Under charging instructions, ES Station 1, with a higher correction coefficient and lower unit cost, exhibits a relatively weaker response, whereas ES Stations 2 and 3 contribute more significantly. ES station 3 provides the strongest charging response due to its lowest response cost. During the fourth phase, Strategies C and D deliver higher power outputs than Strategy B. Although Strategy B achieves the lowest total response cost, which is 7.40 % and 8.73 % lower than Strategies C and D respectively, it also results in significantly lower power allocation to both PS and FR instructions. Specifically, its PS and FR outputs are reduced by 30.94 % and 3.74 % compared to Strategy C, and by 36.27 % and 4.79 % compared to Strategy D. This reduction leads to the largest deviation in joint response output. As shown in Table 4, Strategy A produces less joint response output than Strategies C and D and incurs higher operating costs. Fig. 6 and Fig. 7 further demonstrates that this outcome occurs because Strategy A frequently dispatches ES Station 1,

Table 4

Joint response results under different strategies.

Strategy	Cumulative PS Response Output/MW	Cumulative FR Response Output/MW	PS Output Deviation	FR Output Deviation	Total Cost/CNY
Strategy A	125.000	2635.039	5.845	2.420	28,277.1
Strategy B	125.000	2540.607	5.845	4.122	25,410.7
Strategy C	180.994	2639.248	4.868	2.158	27,440.6
Strategy D	196.118	2668.384	4.587	0.945	27,842.8

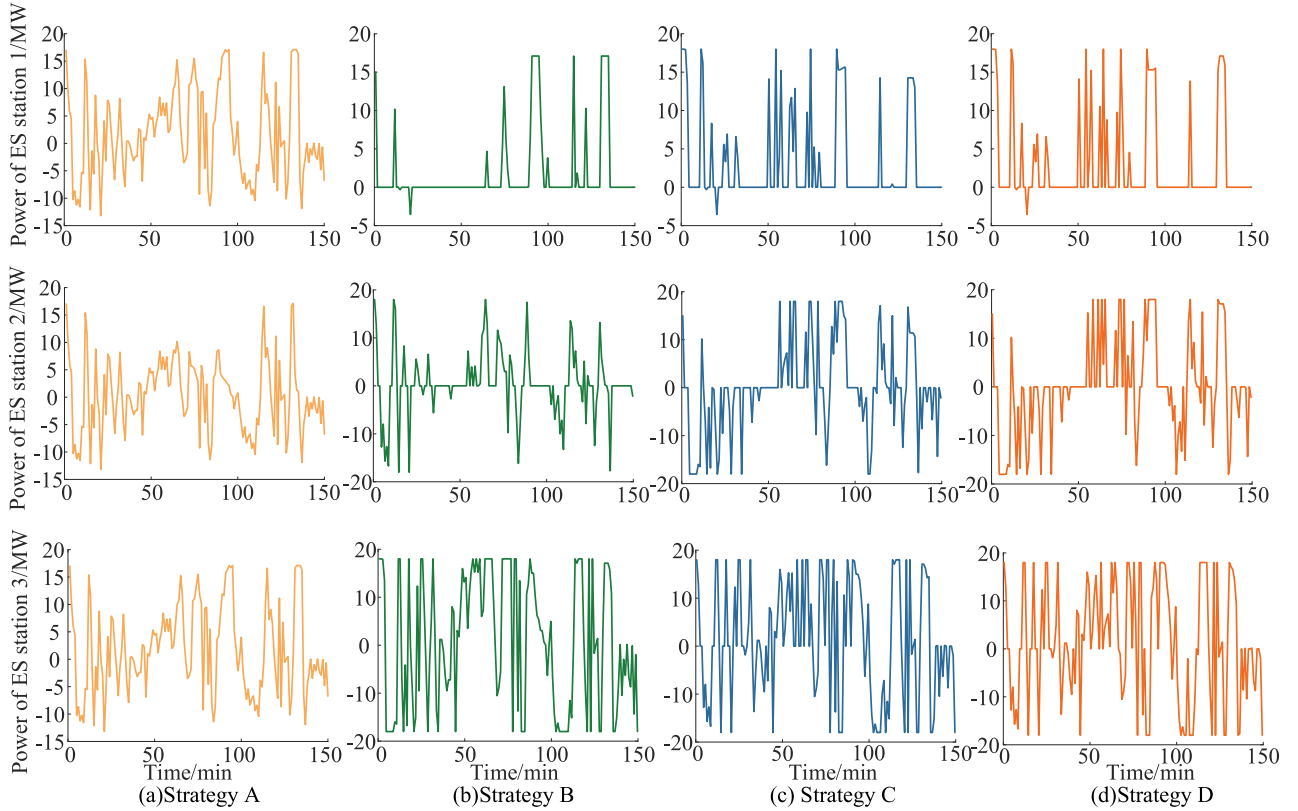


Fig. 6. Joint response output of ES stations cluster under different strategies.

which incurs relatively higher response costs, without accounting for the residual response capacities of the other stations. Consequently, compared with Strategies C and D, which incorporate dispatchable capacity correction coefficients, Strategy A produces suboptimal results in terms of both response effectiveness and economic efficiency.

In summary, while Strategy B performs better in terms of cost, it sacrifices accuracy in response. In contrast, although Strategy A achieves a higher response output than Strategy B, its excessively high operating costs render it economically infeasible. By incorporating dispatchable capacity correction coefficients, Strategies C and D achieve a more balanced task distribution across ES stations. This enhances both the accuracy and reliability of system responses, confirming the effectiveness of the proposed correction mechanism in improving both economic and operational performance.

2) Impact of SOC Optimization on ES Recovery Performance

As shown in Table 5 and Fig. 8, Strategy D demonstrates enhanced performance by incorporating a SOC allocation mechanism, which enables dynamic adjustment of charging and discharging actions according to the real-time SOC levels of individual energy storage units. Units with relatively high SOC are prioritized for discharging, while those with lower SOC are assigned charging tasks. This targeted allocation leads to a more balanced utilization of available capacity and strengthens the system's ability to sustain responses during periods of high demand.

According to the results presented in Table 5, Strategy D achieves the highest overall SOC recovery, exceeding that of Strategy A by 12.87 %, Strategy B by 24.82 %, and Strategy C by 4.36 %. From Fig. 7 and Fig. 8(a), the joint response output under Strategy A is distributed relatively evenly among the three ES stations, which is detrimental to their SOC recovery. Furthermore, Fig. 7 and Fig. 8(b) show that Strategy B, which aims to minimize operating costs, dispatches only the lowest-cost station 3 until its output becomes insufficient. This dispatching strategy does not facilitate SOC recovery at stations 1 and 2, which exhibit relatively

low SOC levels. As shown in Fig. 8(c), Strategy C, which lacks SOC coordination, produces a nearly uniform and suboptimal SOC recovery pattern across all units. For example, ES units 2 and 4 in station 2 exhibit substantially lower SOC recovery than under Strategy D, whereas unit 6 in station 1 reaches its minimum SOC earlier as a result of uncoordinated discharging. By contrast, As shown in Fig. 8(d), Strategy D effectively allocates charging power to units with insufficient SOC, resulting in substantial recovery. Specifically, ES station 2 receives prioritized charging when SOC is low, with units 1, 2, 4, and 5 showing notable improvements. Meanwhile, unit 6 with a higher SOC demonstrates limited recovery, reflecting the mechanism's ability to avoid unnecessary charging.

In summary, the strategy proposed in this study demonstrates significant economic and technical feasibility for deploying ES in providing PS and FR auxiliary services under current environmental and resource conditions. Therefore, the strategy presents promising prospects for practical application.

5.3.2. Comparative validation and performance assessment of the interval SOC recovery method

Building on the SOC interval recovery method proposed in Section 3.3, this section conducts a comparative validation against the conventional approach in [21], which sets a single ideal SOC recovery value at 0.5. Relying on the two-stage optimization strategy developed in this paper, the two methods are evaluated in terms of response output deviation, total operating cost, and SOC recovery deviation. For clarity, the interval recovery method is denoted as Method I, while the single ideal recovery value method is denoted as Method II.

As shown in Table 6, Method I significantly outperforms Method II in terms of cumulative response deviation, with reductions of approximately 7.71 % in PS Output Deviation and 4.38 % in FR Output Deviation. In terms of total operating cost, Method I shows an increase of about 0.48 % compared with Method II. However, this is attributable to

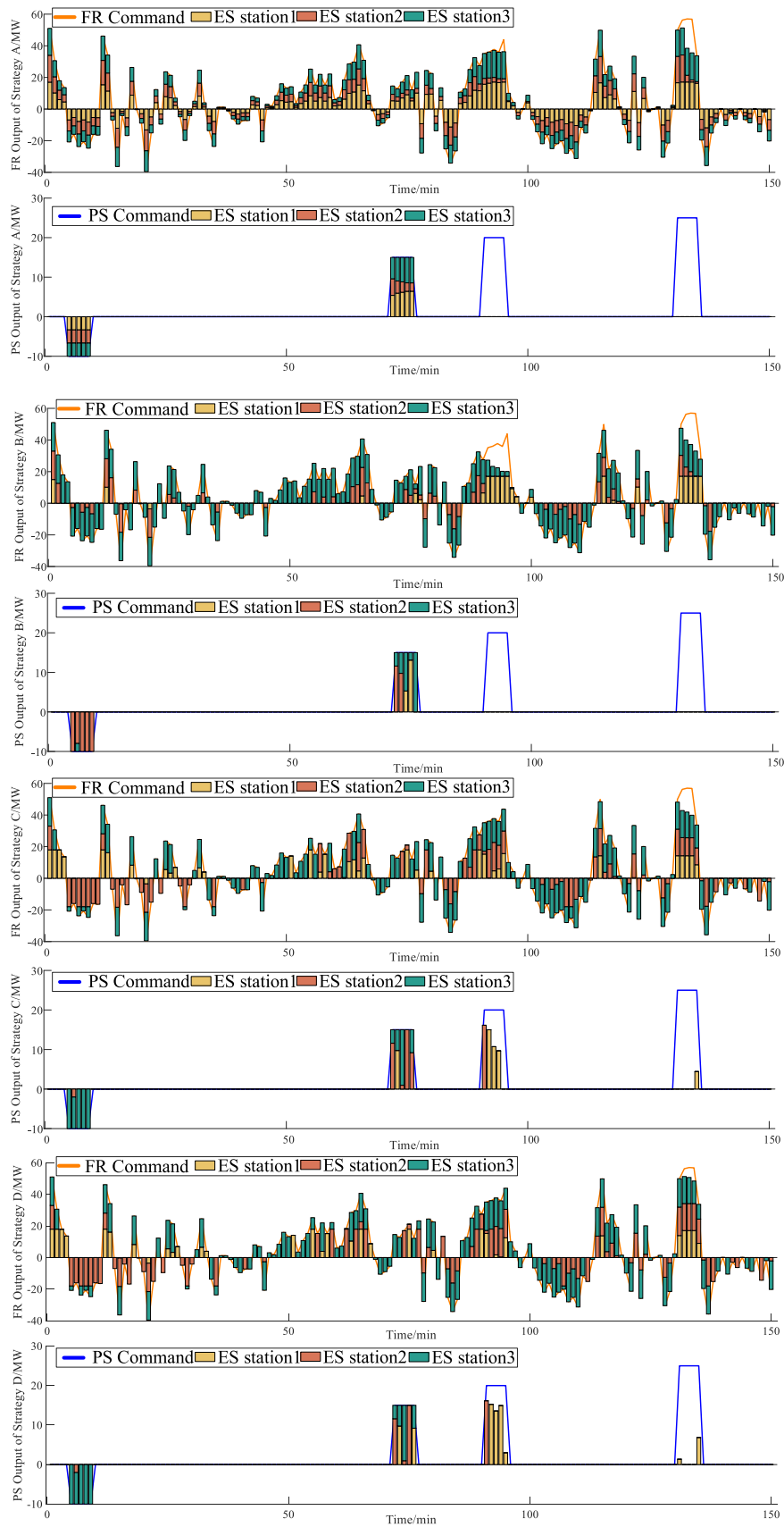


Fig. 7. Composition of ES stations cluster response outputs under different strategies.

Table 5
SOC recovery deviation metrics for ES units under different strategies.

Strategy	Station	SOC Recovery Deviation	Total SOC Recovery Deviation
Strategy A	ES Station 1	0.061	0.272
	ES Station 2	0.143	
	ES Station 3	0.068	
Strategy B	ES Station 1	0.066	0.307
	ES Station 2	0.144	
	ES Station 3	0.097	
Strategy C	ES Station 1	0.060	0.244
	ES Station 2	0.094	
	ES Station 3	0.089	
Strategy D	ES Station 1	0.049	0.233
	ES Station 2	0.092	
	ES Station 3	0.092	

its higher response output, which in turn yields improved response performance. Regarding SOC recovery, Method I reduces the Total SOC Recovery Deviation by about 36.34 % relative to Method II, demonstrating a stronger recovery capability.

Therefore, the comparative results indicate that the interval recovery method consistently outperforms the conventional single-value approach in response deviation, operating cost, and SOC recovery deviation, thereby achieving a markedly improved optimization outcome. This confirms that the proposed interval SOC recovery method offers strong potential in both optimization performance and practical applications.

5.3.3. Extension verification for large-scale power grids

With the growing demand for regional joint responses, this study expands the configuration to twelve ES stations to evaluate the applicability of the proposed two-stage optimization strategy under large-scale power grid conditions. The expanded ES station cluster adopts the same parameter settings as the original three ES stations scenario, thereby substantially increasing the total capacity and output power to more accurately reflect the characteristics of large-scale power grid operations.

Based on the proposed strategy, the optimization results of the scenario with twelve ES stations indicate that the system maintains stability after scaling up, while the optimization model can still be solved within a reasonable computational time. These results further confirm the feasibility of applying the proposed method in transitioning from medium- and small-scale ES systems to large-scale grid scenarios.

To provide a clearer comparison, the specific scenario configurations and operational characteristics are summarized in Table 7, which contrasts the cases with three ES stations and twelve ES stations in terms of station number, total capacity, total output, and optimization computation time. These results demonstrate that the proposed two-stage optimization strategy is not only suitable for small-scale ES systems but also capable of ensuring stability and feasibility in large-scale grid scenarios. This highlights its potential to offer valuable methodological guidance for the scheduling and optimization of future large-scale ES deployments.

Table 6
Joint response results under different strategies.

Method	PS Output Deviation	FR Output Deviation	Total SOC Recovery Deviation	Total Cost/CNY
Method I	4.587	0.945	0.233	27,842.8
Method II	4.797	1.024	0.366	27,708.9

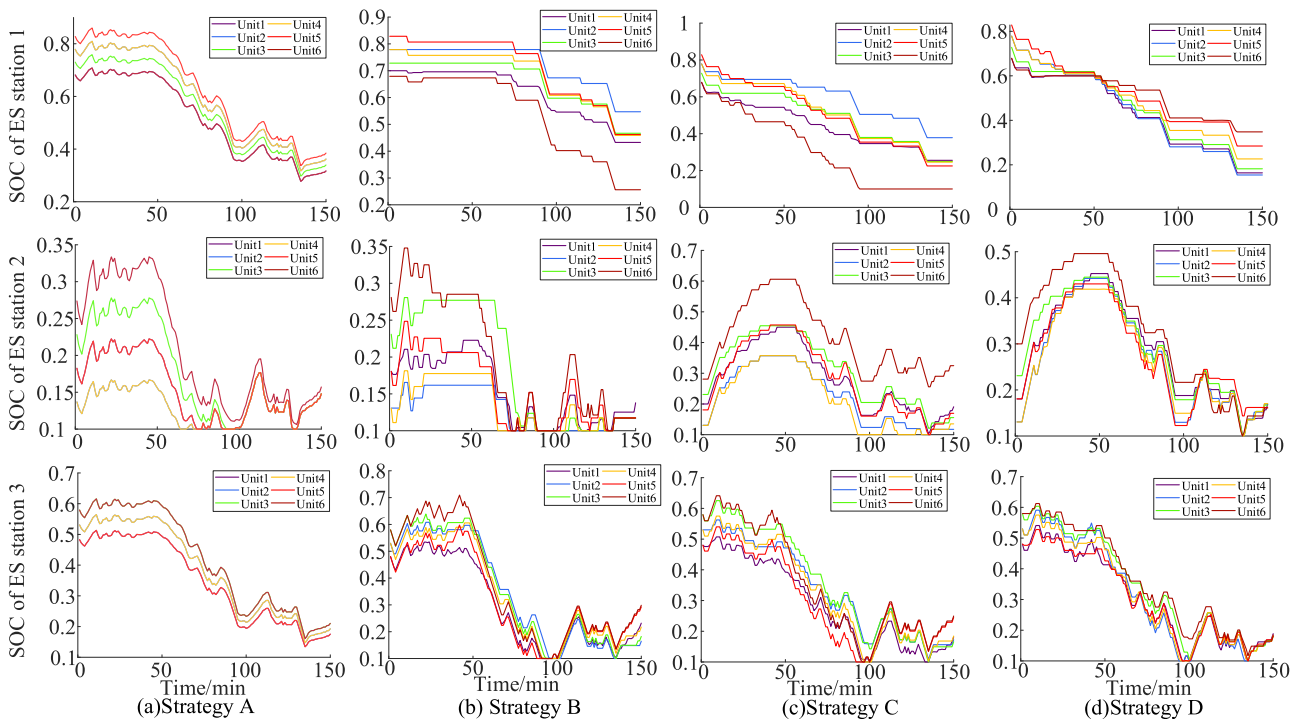


Fig. 8. The SOC curves of ES units in each ES station under different strategies.

Table 7
Comparative results of different ES station scales.

Scenario	Number of ES stations	Total Capacity /MWh	Total Power /MW	Optimization runtime/s
Small-scale scenario	3	54	54	108.5264
Large-scale scenario	12	216	216	169.0565

6. Conclusion

This paper proposes a joint PS and FR strategy for ES stations cluster considering battery dispatchable capacity. The strategy minimizes the operational cost of regional ES stations participating in PS and FR while ensuring the recovery of the SOC for each storage unit. A staged modeling and power allocation approach is developed to coordinate the joint response of multiple regional ES stations to PS and FR demands. Simulation results demonstrate the effectiveness of the proposed strategy, leading to the following conclusions:

- 1) The proposed strategy effectively balances cost and responsiveness for PS and FR, achieving superior economic and technical performance. Compared with other strategies, it significantly reduces deviations in both PS and FR responses, thereby improving the system's overall performance in coordinated dispatch.
- 2) The optimization model incorporates a dispatchable capacity correction coefficient, which enhances the allocation of joint response output among ES stations. ES stations with lower operational costs and stronger response capabilities are prioritized and undertake a larger share of the joint task, while those with relatively higher costs participate when their available capacity is sufficient. This targeted allocation improves overall tracking performance and ensures efficient utilization of regional ES resources.
- 3) The proposed strategy shows distinct advantages in the unit stage. It assigns units with high SOC for discharging and those with low SOC for charging, ensuring rational energy allocation and enhancing SOC recovery. Consequently, the system achieves better long-term regulation and promotes balanced utilization of the ES units.

Further research is essential to improve coordinated dispatch of PS and FR across multiple ES systems and address the complex, dynamic demands for ancillary services. Designing and optimizing cross-regional ES cooperative regulation mechanisms remains critical.

CRediT authorship contribution statement

Feixiang Qiu: Writing – original draft, Software, Investigation, Formal analysis. **Zhihao Yang:** Writing – review & editing, Supervision. **Shaojie Zhu:** Project administration, Data curation. **Zhujun Zhu:** Methodology, Investigation. **Haoming Liu:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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